

Lab Report

Course Title: Computer Networks Laboratory
Course Code: CSE-3634

Spring-2023

Lab No: 8

Name of Labwork: GNS3 installation and configuring a simple network.

TEAM_DIVERGENT

Student's ID : C201249
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Submitted To : Mr. AbdullahilKafi
Assistant Professor

Mark :

1. Introduction:

The experiment is conducted using GNS3, a network simulation software. In this lab experiment, we explore IP connectivity within a network consisting of two LANs: **LAN-A** and **LAN-B**. Each LAN is connected to a switch, and PCs are attached to these switches, creating distinct subnetworks. The objective is to test IP communication between PCs in separate LANs, using GNS3 simulation software. Through ping tests, we examine the practical aspects of IP connectivity and gain insights into network configuration. This experiment offers hands-on experience and highlights the significance of IP connectivity in real-world networks. The findings contribute to a better understanding of network design and administration.

2. Description:

This lab experiment delves into the realm of network communication by creating a practical scenario where two Local Area Networks (LANs), denoted as LAN-A and LAN-B, are interconnected. Within each LAN, a set of PCs is connected to respective switches, forming distinct subnetworks. The primary objective of this experiment is to examine the efficacy of IP communication between PCs situated in separate LANs and to understand the underlying principles governing inter-LAN connectivity.

By configuring and analyzing this two-LAN network model using the GNS3 simulation software, we aim to explore the nuances of IP addressing, routing, and switch configuration that enable or impede successful inter-LAN communication. Through a series of ping tests between PCs, we endeavor to shed light on the complexities and practical considerations involved in establishing robust IP connectivity across distinct network segments. This experiment not only provides hands-on experience in network configuration but also offers valuable insights into the importance of IP connectivity in real-world network scenarios. The findings and observations presented herein serve as a foundation for a deeper understanding of network design and administration principles.

3. Module Overview:

In this module, we delve into the practical application of network concepts using the GNS3 simulation software. We focus on creating a two-LAN network model, with each LAN hosting a set of PCs connected to switches. The module provides a hands-on experience in configuring switches, setting up IP addresses, and testing inter-LAN communication through ping tests. As we progress, we'll explore the intricacies of network design, routing, and troubleshooting. By the end of this module, you'll have gained valuable insights into the fundamentals of IP connectivity and its significance in real-world network scenarios. These skills are essential for network administrators and engineers alike, making this module a crucial step in building networking expertise.

4. Setting Ip of the PCs:

PC1:

A screenshot of a GNS3 terminal window for PC1. The terminal shows the configuration of PC1 with the following details: NAME: PC1[1], IP/MASK: 0.0.0.0/0, GATEWAY: 0.0.0.0, DNS: 00:50:79:66:68:07, MAC: 10032, LPORT: 10032, RHOST:PORT: 127.0.0.1:10033, and HTU: 1500. Below this, the command 'ip 198.168.1.1' is entered, followed by a check for duplicate addresses. The output shows 'PC1 : 198.168.1.1 255.255.255.0'. The terminal window has a title bar with 'solarwinds Solar-PuTTY free tool' and a copyright notice for 2019 SolarWinds Worldwide, LLC.

```
PC1> show ip
NAME       : PC1[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        : 00:50:79:66:68:07
MAC        : 10032
LPORT      : 10032
RHOST:PORT : 127.0.0.1:10033
HTU        : 1500

PC1> ip 198.168.1.1
Checking for duplicate address...
PC1 : 198.168.1.1 255.255.255.0

PC1>
```

PC2:

A screenshot of a GNS3 terminal window for PC2. The terminal shows the configuration of PC2 with the following details: NAME: PC2[1], IP/MASK: 0.0.0.0/0, GATEWAY: 0.0.0.0, DNS: 00:50:79:66:68:00, MAC: 10018, LPORT: 10018, RHOST:PORT: 127.0.0.1:10019, and HTU: 1500. Below this, the command 'ip 198.168.1.2' is entered, followed by a check for duplicate addresses. The output shows 'PC1 : 198.168.1.2 255.255.255.0'. The terminal window has a title bar with 'solarwinds Solar-PuTTY free tool' and a copyright notice for 2019 SolarWinds Worldwide, LLC.

```
PC2> show ip
NAME       : PC2[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        : 00:50:79:66:68:00
MAC        : 10018
LPORT      : 10018
RHOST:PORT : 127.0.0.1:10019
HTU        : 1500

PC2> ip 198.168.1.2
Checking for duplicate address...
PC1 : 198.168.1.2 255.255.255.0

PC2>
```

PC3:

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Dailing.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> ip 198.168.1.3
Checking for duplicate address...
PC1 : 198.168.1.3 255.255.255.0

PC3>
```

PC4:

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Dailing.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> ip 198.168.1.4
Checking for duplicate address...
PC1 : 198.168.1.4 255.255.255.0

PC4>
```

PC5:

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Dailing.
Build time: Apr 10 2019 02:42:20
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC5> ip 198.168.1.5
Checking for duplicate address...
PC1 : 198.168.1.5 255.255.255.0

PC5>
```

PC6:

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Dailing.
Build time: Apr 10 2019 02:42:20
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC6> ip 198.168.1.6
Checking for duplicate address...
PC1 : 198.168.1.6 255.255.255.0

PC6>
```

PC7:

```
PC7> show ip

NAME       : PC7[1]
IP/MASK     : 0.0.0.0/0
GATEWAY     : 0.0.0.0
DNS         :
HWADDR     : 00:50:79:66:68:06
LPORT      : 10030
RHOST:PORT  : 127.0.0.1:10031
RTTU       : 1500

PC7> ip 198.168.1.7
Checking for duplicate address...
PC1 : 198.168.1.7 255.255.255.0

PC7>
```

PC8:

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Dailing.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC8> ip 198.168.1.8
Checking for duplicate address...
PC1 : 198.168.1.8 255.255.255.0

PC8>
```

5. Ping Connectivity:

To verify the connectivity and assess the network link's functionality, a ping test is conducted between all the devices.

In the conducted series of ping tests, PC1 successfully established IP connectivity with all the tested PCs within the network, including **PC2, PC3, PC4, PC5, PC6, PC7, and PC8**. Each ping request yielded timely responses, with response times measured in milliseconds (*ms*). These results confirm the effective IP communication between PC1 and the other networked devices. The response times observed indicate low network latency, ensuring efficient data transfer and network performance. This successful outcome underscores the functionality of the network configuration and routing, facilitating seamless communication between devices across the network.

Code of testing ping in PC1 console:

```
PC1> ping 198.168.1.2
84 bytes from 198.168.1.2 icmp_seq=1 ttl=64 time=1.097 ms
84 bytes from 198.168.1.2 icmp_seq=2 ttl=64 time=1.016 ms
84 bytes from 198.168.1.2 icmp_seq=3 ttl=64 time=2.066 ms
84 bytes from 198.168.1.2 icmp_seq=4 ttl=64 time=1.518 ms
84 bytes from 198.168.1.2 icmp_seq=5 ttl=64 time=1.144 ms
```

```
PC1> ping 198.168.1.3
84 bytes from 198.168.1.3 icmp_seq=1 ttl=64 time=2.265 ms
84 bytes from 198.168.1.3 icmp_seq=2 ttl=64 time=1.019 ms
84 bytes from 198.168.1.3 icmp_seq=3 ttl=64 time=0.917 ms
84 bytes from 198.168.1.3 icmp_seq=4 ttl=64 time=1.136 ms
84 bytes from 198.168.1.3 icmp_seq=5 ttl=64 time=1.356 ms
```

```
PC1> ping 198.168.1.4
84 bytes from 198.168.1.4 icmp_seq=1 ttl=64 time=0.864 ms
84 bytes from 198.168.1.4 icmp_seq=2 ttl=64 time=1.140 ms
84 bytes from 198.168.1.4 icmp_seq=3 ttl=64 time=1.362 ms
84 bytes from 198.168.1.4 icmp_seq=4 ttl=64 time=2.175 ms
84 bytes from 198.168.1.4 icmp_seq=5 ttl=64 time=1.041 ms
```

```
PC1> ping 198.168.1.5
84 bytes from 198.168.1.5 icmp_seq=1 ttl=64 time=1.007 ms
84 bytes from 198.168.1.5 icmp_seq=2 ttl=64 time=1.289 ms
84 bytes from 198.168.1.5 icmp_seq=3 ttl=64 time=2.475 ms
84 bytes from 198.168.1.5 icmp_seq=4 ttl=64 time=1.203 ms
84 bytes from 198.168.1.5 icmp_seq=5 ttl=64 time=1.822 ms
```

```
PC1> ping 198.168.1.6
84 bytes from 198.168.1.6 icmp_seq=1 ttl=64 time=0.892 ms
84 bytes from 198.168.1.6 icmp_seq=2 ttl=64 time=1.085 ms
84 bytes from 198.168.1.6 icmp_seq=3 ttl=64 time=0.949 ms
84 bytes from 198.168.1.6 icmp_seq=4 ttl=64 time=2.108 ms
84 bytes from 198.168.1.6 icmp_seq=5 ttl=64 time=1.290 ms
```

```
PC1> ping 198.168.1.7
84 bytes from 198.168.1.7 icmp_seq=1 ttl=64 time=0.816 ms
84 bytes from 198.168.1.7 icmp_seq=2 ttl=64 time=2.429 ms
84 bytes from 198.168.1.7 icmp_seq=3 ttl=64 time=1.635 ms
84 bytes from 198.168.1.7 icmp_seq=4 ttl=64 time=0.723 ms
84 bytes from 198.168.1.7 icmp_seq=5 ttl=64 time=0.923 ms
```

```
PC1> ping 198.168.1.8
84 bytes from 198.168.1.8 icmp_seq=1 ttl=64 time=0.941 ms
84 bytes from 198.168.1.8 icmp_seq=2 ttl=64 time=0.666 ms
84 bytes from 198.168.1.8 icmp_seq=3 ttl=64 time=1.482 ms
84 bytes from 198.168.1.8 icmp_seq=4 ttl=64 time=1.676 ms
84 bytes from 198.168.1.8 icmp_seq=5 ttl=64 time=0.960 ms
```

```

PC1> ping 198.168.1.2
84 bytes from 198.168.1.2 icmp_seq=1 ttl=64 time=1.097 ms
84 bytes from 198.168.1.2 icmp_seq=2 ttl=64 time=1.016 ms
84 bytes from 198.168.1.2 icmp_seq=3 ttl=64 time=2.066 ms
84 bytes from 198.168.1.2 icmp_seq=4 ttl=64 time=1.518 ms
84 bytes from 198.168.1.2 icmp_seq=5 ttl=64 time=1.144 ms

PC1> ping 198.168.1.3
84 bytes from 198.168.1.3 icmp_seq=1 ttl=64 time=2.265 ms
84 bytes from 198.168.1.3 icmp_seq=2 ttl=64 time=1.019 ms
84 bytes from 198.168.1.3 icmp_seq=3 ttl=64 time=0.917 ms
84 bytes from 198.168.1.3 icmp_seq=4 ttl=64 time=1.136 ms
84 bytes from 198.168.1.3 icmp_seq=5 ttl=64 time=1.356 ms

PC1> ping 198.168.1.4
84 bytes from 198.168.1.4 icmp_seq=1 ttl=64 time=0.864 ms
84 bytes from 198.168.1.4 icmp_seq=2 ttl=64 time=1.140 ms
84 bytes from 198.168.1.4 icmp_seq=3 ttl=64 time=1.362 ms
84 bytes from 198.168.1.4 icmp_seq=4 ttl=64 time=2.175 ms
84 bytes from 198.168.1.4 icmp_seq=5 ttl=64 time=1.041 ms

PC1> ping 198.168.1.5
84 bytes from 198.168.1.5 icmp_seq=1 ttl=64 time=1.007 ms
84 bytes from 198.168.1.5 icmp_seq=2 ttl=64 time=1.289 ms
84 bytes from 198.168.1.5 icmp_seq=3 ttl=64 time=2.475 ms
84 bytes from 198.168.1.5 icmp_seq=4 ttl=64 time=1.203 ms
84 bytes from 198.168.1.5 icmp_seq=5 ttl=64 time=1.822 ms

PC1> ping 198.168.1.6
84 bytes from 198.168.1.6 icmp_seq=1 ttl=64 time=0.892 ms
84 bytes from 198.168.1.6 icmp_seq=2 ttl=64 time=1.085 ms
84 bytes from 198.168.1.6 icmp_seq=3 ttl=64 time=0.949 ms
84 bytes from 198.168.1.6 icmp_seq=4 ttl=64 time=2.108 ms
84 bytes from 198.168.1.6 icmp_seq=5 ttl=64 time=1.290 ms

PC1> ping 198.168.1.7
84 bytes from 198.168.1.7 icmp_seq=1 ttl=64 time=0.816 ms
84 bytes from 198.168.1.7 icmp_seq=2 ttl=64 time=2.429 ms
84 bytes from 198.168.1.7 icmp_seq=3 ttl=64 time=1.635 ms
84 bytes from 198.168.1.7 icmp_seq=4 ttl=64 time=0.723 ms
84 bytes from 198.168.1.7 icmp_seq=5 ttl=64 time=0.923 ms

PC1> ping 198.168.1.8
84 bytes from 198.168.1.8 icmp_seq=1 ttl=64 time=0.941 ms
84 bytes from 198.168.1.8 icmp_seq=2 ttl=64 time=0.666 ms
84 bytes from 198.168.1.8 icmp_seq=3 ttl=64 time=1.402 ms
84 bytes from 198.168.1.8 icmp_seq=4 ttl=64 time=1.676 ms
84 bytes from 198.168.1.8 icmp_seq=5 ttl=64 time=0.960 ms

PC1>

```

Fig 9: verify the connectivity in PC1 console

9. Build and Simulation:

This simulation employs GNS3, a network simulation software, to create a practical network environment with two LANs (LAN-A and LAN-B). Each LAN is equipped with switches and PCs. The goal is to test IP connectivity between PCs in different LANs. This hands-on experience explores the fundamentals of IP addressing, routing, and switch configuration, offering valuable insights into real-world network scenarios.

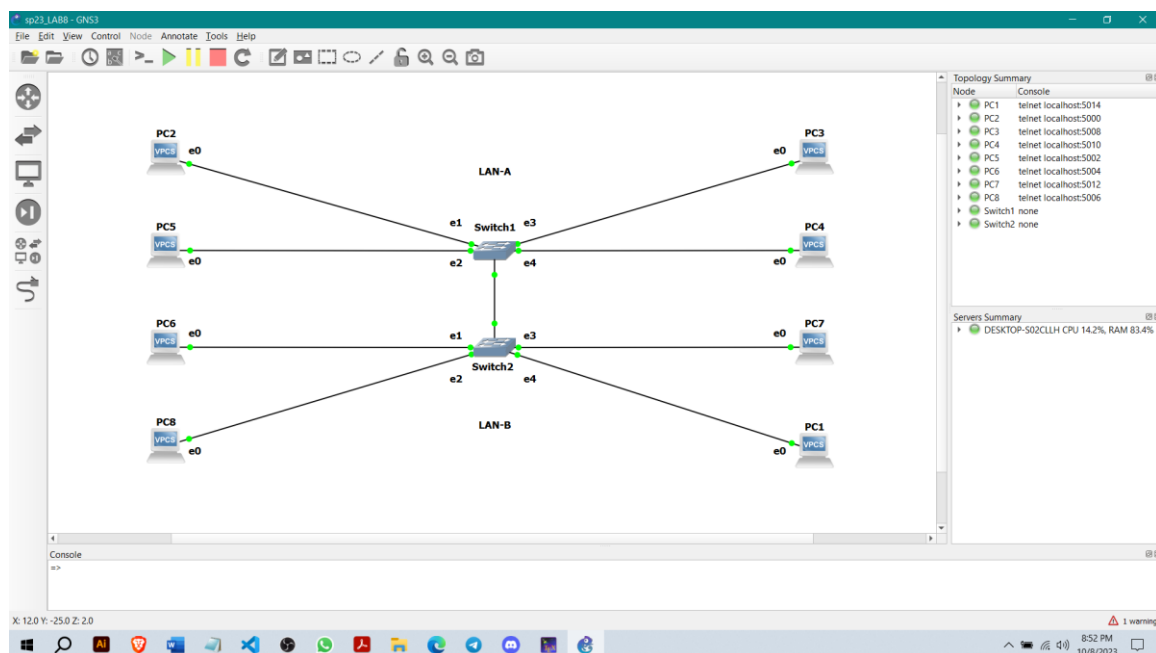


Fig 10: Simulation of Lab6

10. Result Analysis:

In our experiment, we observed that PCs within the same LAN demonstrated smooth IP communication due to their shared network segment, while successful communication between PCs in different LANs required routing mechanisms facilitated by properly configured routers. The effectiveness of inter-LAN communication depended on accurate IP addressing, including subnetting, as well as correct routing table configurations. This experiment also highlighted potential challenges in network administration, such as IP conflicts and misconfigurations, underscoring the importance of meticulous network design and troubleshooting skills. In conclusion, our findings emphasize the vital role of IP addressing, routing, and switch configuration in enabling IP connectivity between separate LANs, with these fundamental principles being pivotal in real-world network scenarios.

11. Conclusion:

In this lab, we explored IP connectivity in a two-LAN setup using GNS3. We found that successful communication between PCs in separate LANs required effective routing configurations. This experiment emphasized the importance of proper network design and troubleshooting in ensuring robust IP connectivity. The skills and insights gained here are foundational for network administrators and engineers, highlighting the critical role of IP connectivity in networking.